

MODULAR ARITHMETIC

THERE ARE NO SECRETS TO
SUCCESS. IT IS THE RESULT OF
PREPARATION, HARD WORK, AND
LEARNING FROM FAILURE.

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Good
Morning



01. Modular Arithmetic
02. Calculate $a^n \bmod m$
03. Count pair with $\text{sum} \bmod = 0$
04. Addition & multiplication rule
05. permutation basics
06. Combination basics
07. Pascal Triangle



$A \% B \rightarrow$ remainder when A is divided by B .

$$\text{Dividend} = \text{divisor} * \text{quo} + \text{rem}$$

$$\text{rem} = \text{dividend} - \underbrace{\text{div} * \text{quo}}$$

largest multiple of divisor

$\leq$ dividend.

01. $10 \% 4 = 2$

02. $13 \% 5 = 3$

03. $-40 \% 7 = -40 - (\text{largest mul of } 7 \leq -40)$

$$\Rightarrow -40 - (-42) \Rightarrow \underline{\underline{2}}$$

04 $-60 \% 9 \Rightarrow -60 - (\text{largest mul of } 9 \leq -60)$

$$= -60 - (-63) = -60 + 63 \Rightarrow 3$$

$x \% m$

min ans

0

max ans

$m-1$

* Why %M ?

→ Restrict your output in a specific range

$$\left. \begin{array}{c} -\infty \\ \vdots \\ \vdots \\ \vdots \\ +\infty \end{array} \right\} \% M \Rightarrow \frac{\text{result}}{0 \text{ to } (m-1)}$$

$$M = \underbrace{10^9 + 7}_{\text{prime no.}} \rightarrow \text{Hashmap implementation}$$

* Modular Arithmetic

01. $(a + b) \% m = (a \% m + b \% m) \% m$

$$a = 13$$

$$b = 9$$

$$m = 5$$

$$(13 + 9) \% 5 = (13 \% 5 + 9 \% 5) \% 5$$

$$22 \% 5 = (3 + 4) \% 5$$

$$\Rightarrow 2 = 2$$

02. $(a * b) \% m = (a \% m * b \% m) \% m$

$$a = 13, b = 9, m = 5$$

$$(13 * 9) \% 5 = (13 \% 5 * 9 \% 5) \% 5$$

$$\Rightarrow 117 \% 5 = (3 * 4) \% 5$$

$$\Rightarrow 2 = 2$$

03. $(a - b) \% m = (a \% m - b \% m) \% m$

$$a = 13, b = 9, m = 5$$

$$(13 - 9) \% 5 = (13 \% 5 - 9 \% 5) \% 5$$

$$4 \% 5 = (3 - 4) \% 5$$

$$= 4 = -1 \% 5$$

$$(a - b) \% m = (a \% m - b \% m) \% m + 0$$

$$= (a \% m - b \% m) \% m + m \% m$$

$$(a - b) \% m = (a \% m - b \% m + m) \% m$$

$$* \quad a=3 \quad b=4 \quad m=5$$

$$(3-4) \% 5 = (3 \% 5 - 4 \% 5 + 5) \% 5$$

$$-1 \% 5 = (3 - 4 + 5) \% 5$$

$$\Rightarrow 4 = 4 \% 5$$

4

$\Rightarrow$

4

$$* \quad a \% m = (((a \% m) \% m) \% m) \% m$$

$$* \quad a^b \% m = ((a \% m)^b \% m)$$

$$* \quad (37^{103} - 1) \% 12$$

$$= (37^{103} \% 12 - 1 \% 12 + 12) \% 12$$

$$\Rightarrow ((37 \% 12)^{103} \% 12 - 1 \% 12 + 12) \% 12$$

$$\Rightarrow ((1)^{103} \% 12 - 1 + 12) \% 12$$

$$\Rightarrow (\cancel{1} - \cancel{1} + 12) \% 12$$

$$= 12 \% 12 \Rightarrow 0$$

* Power $\Rightarrow a^n \% m$

$$1 \leq a \leq 10^9$$

$$1 \leq n \leq 10^5$$

$$1 \leq m \leq 10^9$$

* Calculate $a^n \% m$

long ans = 1

for (i = 1; i ≤ n; i++) {

ans = (ans % m * a % m) % m
OR
ans = (ans * a) % m
}

$$n = 3$$

$$a = 10^9$$

$$ans = 1$$

$$i = 1$$

$$ans = 1 * 10^9 = 10^9 \% m = 10^9$$

$$= 10^9 * 10^9 = 10^{18} \% m = 10^9$$

$$= 10^9 * 10^9 = 10^{18} \% m = 10^9$$

```

* int power (a, n, m)
  |
  | if (n==0) return 1;
  |
  | long ans = power (a, n-1, m)
  |
  | return int (ans * a) % m
  |
  3

```

* Fast power function

$$2^8 = 2^4 * 2^4$$

$$2^9 = 2^4 * 2^4 * 2$$

```

int power (a, n, m)
  |
  | if (n==0) return 1
  |
  | long x = power (a, n/2, m);
  |
  | if (n % 2 == 0) {
  |   |
  |   | return (x * x) % m
  |   |   (10^9 * 10^9)
  |   |   10^18 % m ⇒ 10^9
  |   |
  |   }
  |   else {
  |     |
  |     | return (x * x * a) % m
  |     |   (10^9 + 10^9 + 10^9)
  |     |   10^27
  |     |
  |     }
  |
  3

```

$$((x * x) \% m * a \% m) \% m$$



$$\begin{array}{l}
 (x * x * a) \% m \\
 \underbrace{\quad \quad \quad}_{(10^9 + 10^9 + 10^9)} \\
 \underbrace{\quad \quad \quad}_{10^{27}}
 \end{array}$$

X

Q Given $ar[]$, m . Calculate no. of pairs (i, j)

such that $(ar[i] + ar[j]) \% m = 0$

Note :- $i \neq j$ & $pair(i, j) = pair(j, i)$

$ar[] = \{ \underset{0}{4} \underset{1}{7} \underset{2}{6} \underset{3}{5} \underset{4}{5} \underset{5}{3} \}$ $m = 3$

i	j	$ar[i]$	$ar[j]$	$(ar[i] + ar[j]) \% m$
0	3	4	5	$(4+5)/3 = 0$
0	4	4	5	$(4+5) \% 3 = 0$
1	3	7	5	$(7+5) \% 3 = 0$
1	4	7	5	$(7+5) \% 3 = 0$
2	5	6	3	$(6+3) \% 3 = 0$

Ans = 5

Brute force $\rightarrow$ Check for every possible pair (i, j)

if $(ar[i] + ar[j]) \% m == 0 \rightarrow \text{count}++;$

TC: $O(n^2)$

SC: $O(1)$

Idea 2

$$(ar[i] + ar[j]) \% m = 0$$

$$\Rightarrow (ar[i] \% m + ar[j] \% m) \% m = 0$$

$$\underbrace{\quad}_{[0 \ m-1]} \quad \underbrace{\quad}_{[0 \ m-1]}$$

$$0 + 0$$

$$1 + (m-1)$$

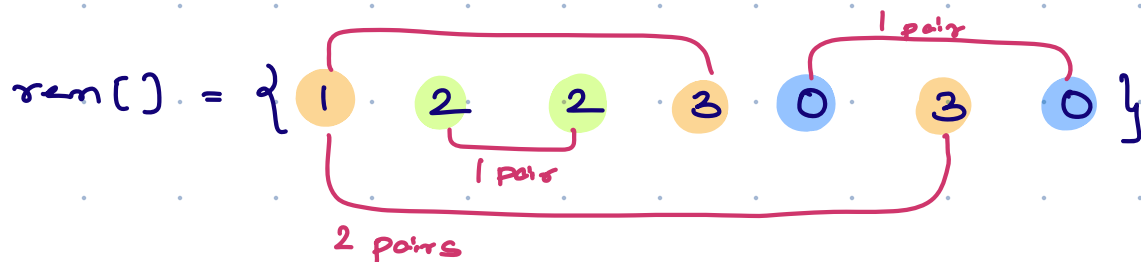
$$2 + (m-2)$$

$$3 + (m-3)$$

⋮

$$k + (m-k)$$

* $ar[] = \{13 \ 14 \ 22 \ 3 \ 32 \ 19 \ 16\} \ m=4$



$$\underline{\underline{Ans = 4}}$$

$$* \text{ ar}[] = \{ \underset{0}{6} \underset{1}{7} \underset{2}{5} \underset{3}{11} \underset{4}{19} \underset{5}{20} \underset{6}{9} \underset{7}{15} \underset{8}{14} \underset{9}{13} \underset{10}{12} \underset{11}{23} \}$$

$m=5$

$$\text{rar}[] = \{ \underset{0}{1} \underset{1}{2} \underset{2}{0} \underset{3}{1} \underset{4}{4} \underset{5}{0} \underset{6}{4} \underset{7}{0} \underset{8}{4} \underset{9}{3} \underset{10}{2} \underset{11}{3} \}$$

$$\text{range of ram} = \{ \underset{0}{3} \underset{1}{2} \underset{2}{2} \underset{3}{2} \underset{4}{3} \}$$

pair

$$1 \& 4 = 2 * 3 = 6$$

$$2 \& 3 = 2 * 2 = 4$$

$$0 \& 0 = {}^3C_2 = 3 \{ (5, 20), (5, 15) \& (20, 15) \}$$



$$\frac{n * (n-1)}{2}$$

$m=10$

$$\text{ar}[] =$$

29	11	21	17	2	5	4	6	23	13	26	14	18	15	30	35	50	20	40	9
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

$$\text{mar} =$$

9	1	1	7	2	5	4	6	3	3	6	4	8	5	0	5	0	0	0	9
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$\text{cnt}[10] =$$

4	2	1	2	2	3	2	1	1	2
0	1	2	3	4	5	6	7	8	9

$$arr[i] \% 10 \quad arr[j] \% 10 = 0$$

$$cnt[0] \longrightarrow {}^4C_2 \text{ pairs}$$

$$cnt[1] * cnt[9] \Rightarrow 4 \text{ pairs}$$

$$cnt[2] * cnt[8] = 1 \text{ pair}$$

$$cnt[3] * cnt[7] = 2 \text{ pairs}$$

$$cnt[4] * cnt[6] = 4 \text{ pairs}$$

$$cnt[5] \longrightarrow {}^3C_2 \text{ pair}$$

```
int countPairs ( arr, n )
```

```
// populate the freq arr
```

```
int [] cnt = new int[m];
```

```
for ( i = 0; i < n; i++ )
```

```
    int rem = arr[i] % m;
    cnt[rem] ++;
```

```
ans = 0
```

```
int a = cnt[0]
```

```
ans = ans + ( a * (a-1) )
                2
```

```
if (m/2 == 0) {
```

```
    int b = cnt[m/2]
```

```
    ans = ans +  $\frac{b * (b-1)}{2}$ 
```

```
}
```

// get all remaining pairs

```
i = 1, j = m-1
```

```
while (i < j) {
```

```
    ans = ans + (cnt[i] * cnt[j]);
```

```
    i++;
```

```
    j--;
```

```
}
```

```
return ans;
```

3

8:40 → 8:50 AM

* Addition & Multiplication Rule

01. Given 3 True/False questions, every question has to be answered. In how many ways can you answer all the questions.

$$\frac{2}{\text{T} \quad \text{F}} * \frac{2}{\text{T} \quad \text{F}} * \frac{2}{\text{T} \quad \text{F}} = 8 \text{ ways}$$

F	F	F
F	F	T
F	T	F
F	T	T
T	F	F
T	F	T
T	T	F
T	T	T

8 ways

Q2 Given 3 girls & 4 boys. How many different pairs can we form?

Note :- pair $\rightarrow$ 1 Girl + 1 Boy

Boys Girl

B_1 G_1

B_2 G_2

B_3 G_3

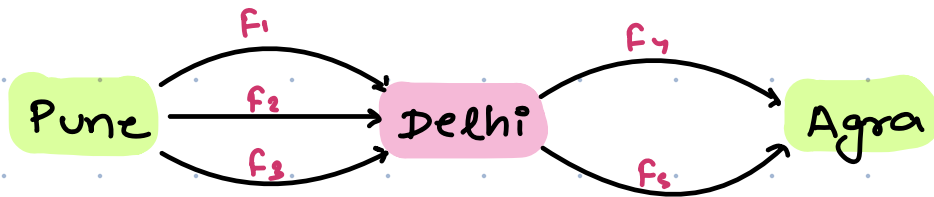
B_4

Ans = 12

1 Boy + 3 Girls

4 Boys * 3 Girls = 12 pairs

Eg 3



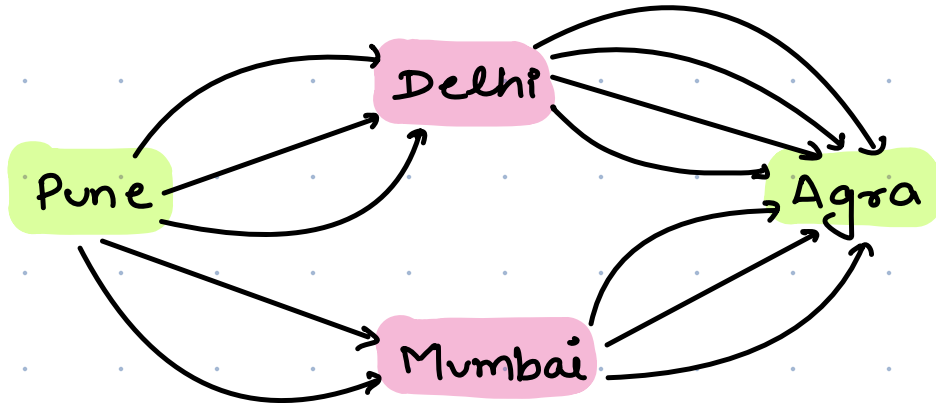
No. of ways to reach Agra from Pune via Delhi

Pune $\rightarrow$ Delhi AND Delhi $\rightarrow$ Agra

$$3 * 2 = 6 \text{ ways}$$

$F_1 F_4$ $F_2 F_4$ $F_3 F_4$
 $F_1 F_5$ $F_2 F_5$ $F_3 F_5$

Eg 4



No. of ways to reach Agra from pune ?

via Delhi

OR

via mumbai?

Pune $\rightarrow$ Delhi AND D $\rightarrow$ A

P $\rightarrow$ M AND M $\rightarrow$ A

$$3 * 4$$

+

$$2 * 3$$

$$= 12 + 6 = 18 \text{ ways}$$

AND $\rightarrow *$

OR $\rightarrow +$

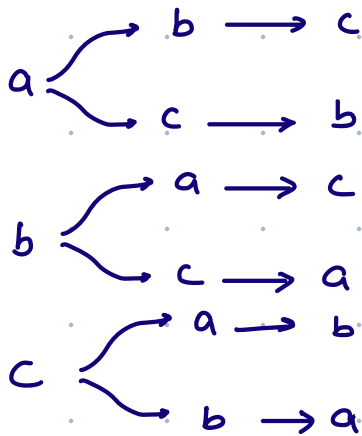
* Permutations $\rightarrow$ No. of arrangements

$(i, j) \neq (j, i) \rightarrow$ order matters

Q1 Given 3 distinct chars. In how many ways we can arrange them?

$S = \{a, b, c\}$

$$\underline{3} * \underline{2} * \underline{1} = \underline{3!} \Rightarrow 6$$



* Arrange n distinct characters $\Rightarrow n!$

Q Given 5 distinct characters, in how many ways you can arrange 2 distinct chars

$$\{a, b, c, d, e\} \quad \underline{5} * \underline{4} \Rightarrow \underline{20} \text{ ways}$$

* N distinct characters, in how many ways you can arrange 3 distinct chars

$$\underline{n * (n-1) * (n-2)}$$

* N distinct characters, in how many ways you can arrange r chars

$$= n * (n-1) * (n-2) * \dots * (n-(r-1))$$

$$= n * (n-1) * (n-2) * \dots * (n-r+1)$$

$$n * (n-1) * (n-2) * \dots * (n-r+1) * (n-r) * (n-r-1) * \dots * 1$$

$$(n-r) * (n-r-1) * \dots * 1$$

$$\frac{n!}{(n-r)!} = {}_n P_r$$

(No. of ways to arrange r places with n distinct characters)